Workflow Automation – Assignment 1: BPMN Modeling

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Repository: https://github.com/us9250-svg/ex1_workflow_automation.git
Branch: main
Tool: Camunda Modeler (v5.49.0) / Camunda 8.9

Overview

This document provides a detailed breakdown of the three Business Process Model and Notation (BPMN) workflows designed for this assignment. All corresponding .bpmn files have been pushed to the linked GitHub repository for review.

❖ Repository Contents

File	Scenario
diagram_1.bpmn	Workflow for Employee Leave Approvals
diagram_2.bpmn	Workflow for Processing Online Purchase Orders
diagram_3.bpmn	Workflow for IT Service Helpdesk Requests

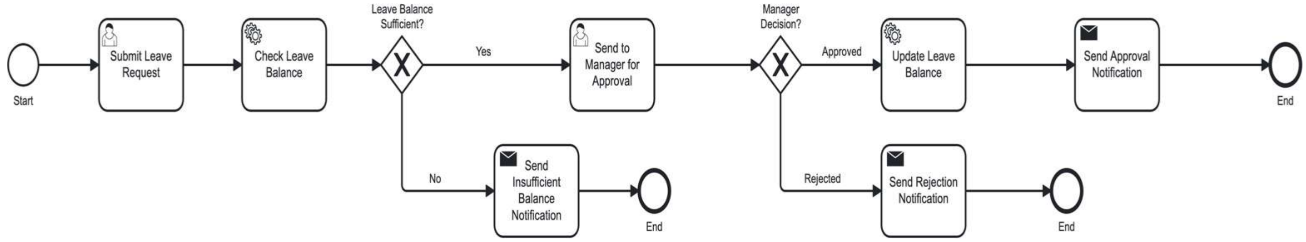
❖ Scenario 1 – Employee Leave Approval

This model maps out how employee leave requests are handled. It kicks off when a user submits their leave application. The HR system automatically evaluates the employee's current leave balance. At the first decision point (Exclusive Gateway), requests with inadequate balances are immediately rejected, triggering a notification before terminating. Conversely, valid balances are routed to a manager for manual review. A second gateway evaluates the manager's verdict: approvals lead to a system update of the leave balance and a success notification, whereas rejections result in a denial message. Both paths conclude the process.

Elements present in diagram_1.bpmn

BPMN Element	Actual element in the uploaded model
Start Event	Start
User Task	Submit Leave Request
Service Task	Check Leave Balance
Exclusive Gateway	Leave Balance Sufficient?
User Task	Send to Manager for Approval
Exclusive Gateway	Manager Decision?
Service Task	Update Leave Balance
Send Task	Send Approval Notification
Send Task	Send Rejection Notification
Send Task	Send Insufficient Balance Notification
End Events	Three End events for the three outcomes

Decision paths: Leave Balance Sufficient = Yes → Send to Manager for Approval. No → Send Insufficient Balance Notification →End. Manager Decision = Approved → Update Leave Balance → Send Approval Notification → End. Rejected → Send Rejection Notification → End.



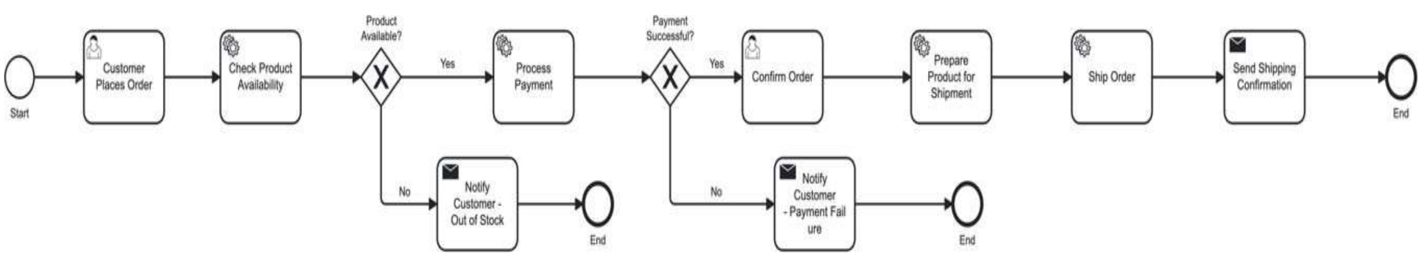
❖ Scenario 2 – Online Purchase Order Processing

This diagram illustrates the backend sequence triggered when a customer buys an item online. Upon order placement, a service task verifies inventory. An exclusive gateway splits the flow based on stock status: out-of-stock items trigger a cancellation notice to the buyer and end the flow. If in stock, the system advances to payment processing. A second gateway validates the transaction. Failed payments notify the customer and halt the sequence. Successful payments trigger a chain of tasks: order confirmation, shipment preparation, dispatching the product, and finally, emailing the shipping details to the user before wrapping up.

Elements present in diagram_2.bpmn

BPMN Element	Actual element in the uploaded model
Start Event	Start
User Task	Customer Places Order
Service Task	Check Product Availability
Exclusive Gateway	Product Available?
Service Task	Process Payment
Exclusive Gateway	Payment Successful?
User Task	Confirm Order
Service Task	Prepare Product for Shipment
Service Task	Ship Order
Send Task	Notify Customer - Out of Stock
Send Task	Notify Customer - Payment Failure
Send Task	Send Shipping Confirmation
End Events	Three End events for completed and failed paths

Decision paths: Product Available = Yes → Process Payment; No → Notify Customer - Out of Stock → End. Payment Successful =Yes → Confirm Order → Prepare Product for Shipment → Ship Order → Send Shipping Confirmation → End. No → Notify Customer - Payment Failure → End.



❖ Scenario 3 – IT Service Request

This model defines the resolution timeline for internal technical issues. The sequence initiates when an employee files a support ticket. The system registers the ticket and assesses its severity. An exclusive gateway then assigns the ticket based on this assessment: critical issues go to a Senior Technician, while minor ones are handled by a Support Technician. Both assignment paths converge at the investigation phase. A subsequent gateway determines if the team can fix the bug in-house. Resolvable issues are fixed internally, while unresolvable ones are escalated to third-party vendors. Finally, the paths merge again to update the ticket status in the system and alert the employee of the resolution before the process terminates.

Elements present in diagram_3.bpmn

BPMN Element	Actual element in the uploaded model
Start Event	Start
User Task	Submit IT Support Request
Service Task	Register Request
Service Task	Check Problem Severity
Exclusive Gateway	Problem Severity?
User Task	Assign to Senior Technician
User Task	Assign to Support Technician
Service Task	Investigate Problem
Exclusive Gateway	Can Be Resolved?
Service Task	Fix Problem
Service Task	Escalate to External Service Provider
Service Task	Update Request Status
Send Task	Send Resolution Notification
End Event	End

Decision paths: Problem Severity = High → Assign to Senior Technician; Low → Assign to Support Technician. Can Be Resolved = Yes → Fix Problem; No → Escalate to External Service Provider. Both paths merge at Update Request Status.